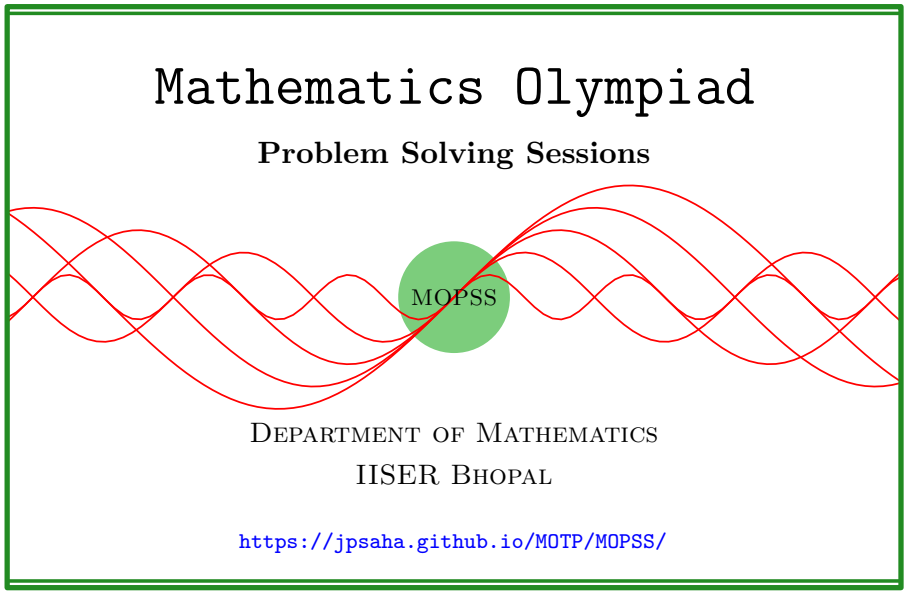
Functions



Suggested readings

- Evan Chen's advice On reading solutions, available at https://blog.evanchen.cc/2017/03/06/on-reading-solutions/.
- Evan Chen's Advice for writing proofs/Remarks on English, available at https://web.evanchen.cc/handouts/english/english.pdf.
- Notes on proofs by Evan Chen from OTIS Excerpts [**Che25**, Chapter 1].
- Tips for writing up solutions by Edward Barbeau, available at https://www.math.utoronto.ca/barbeau/writingup.pdf.
- Evan Chen discusses why math olympiads are a valuable experience for high schoolers in the post on Lessons from math olympiads, available at https://blog.evanchen.cc/2018/01/05/lessons-from-math-olympiads/.

List of problems and examples

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§1 Functions

Exercise 1 (All-Russian Mathematical Olympiad 2014 Grade 10 Day 1 P2, , by O. Podlipsky)**.** Let $f : \mathbb{R} \to \mathbb{R}$ be a function such that

$$f(x)^2 \leq f(y)$$

holds for any $x, y \in \mathbb{R}$ with $x > y$. Show that

$$0 \leq f(x) \leq 1$$

holds for any $x \in \mathbb{R}$.

Walkthrough —

(a) Show that $f(x) \geq 0$ for any $x \in \mathbb{R}$.

(b) Using the given inequality, show that for any $x, y \in \mathbb{R}$ with $x > y$, and for any positive integer n, the inequality

$$f(x)^{2^n} \leq f(y)$$

holds.

(c) Use the above to conclude that $f(x) \leq 1$ for any $x \in \mathbb{R}$.

Solution 1. Note that for any $x \in \mathbb{R}$, we have

$$f(x+1)^2 \leq f(x),$$

which shows that $f(x) \geq 0$.

To show that $f(x) \leq 1$ for any $x \in \mathbb{R}$, let us establish the following claim.

Claim — For any $x, y \in \mathbb{R}$ with $x > y$, and for any positive integer n, the inequality

$$f(x)^{2^n} \leq f(y)$$

holds.

Proof of the Claim. Note that the inequality holds for $n = 1$ by hypothesis. Let us assume that n is a positive integer such that the inequality holds for any $x, y \in \mathbb{R}$ with $x > y$. Then, for any $x, y \in \mathbb{R}$ with $x > y$, we have

$$x > \frac{x+y}{2} > y,$$

which implies that

$$f(x)^{2^n} \leq f\left(\frac{x+y}{2}\right) \quad \text{and} \quad f\left(\frac{x+y}{2}\right)^2 \leq f(y).$$

Combining these two inequalities and using that $f(x)$ is nonnegative, we obtain

$$f(x)^{2^{n+1}} \leq f(y).$$

By the principle of mathematical induction, the claim follows. □

Let x be an arbitrary real number. Using the above claim, we obtain

$$f(x)^{2^n} \leq f(x-1)$$

for **any** positive integer n. If $f(x) \leq 1$, then we are done. It remains to consider the case that $f(x) > 1$, which we assume from now on. It follows that

$$f(x-1) \geq 1 + 2^n \left(f(x) - 1\right)$$

for **any** positive integer n. This is impossible since $f(x) - 1$ is positive.

Combining the above, we conclude that

$$0 \leq f(x) \leq 1$$

holds for any $x \in \mathbb{R}$. ■

Exercise 2 (INMO 2026 P2,). Let $f : \mathbb{N} \to \mathbb{N}$ be a function satisfying the following condition: for each $k > 2026$, the number $f(k)$ equals the maximum number of times a number appears in the list $f(1), f(2), \ldots, f(k-1)$. Prove that $f(n) = f(n + f(n))$ for infinitely many $n \in \mathbb{N}$. (Here $\mathbb{N}$ denotes the set $\{1, 2, 3, \ldots\}$ of positive integers.)

Walkthrough —

(a) Show that $f|_{\{2027,2028,\ldots\}}$ is a non-decreasing function.

(b) Show that f takes arbitrarily large values.

(c) Show that there exists an integer $r \geq 2027$ such that

$$f(r) > \max\{f(1), f(2), \ldots, f(r-1)\}.$$

(d) Show that for such an integer r, the values of f at the integers $r, r+1, r+$

$2, \ldots, r + f(r)$ are all equal, and $f(r + f(r) + 1) = f(r) + 1$.

(e) Conclude the solution.

Solution 2. Note that $f(n) \leq f(n+1)$ holds for any $n \geq 2027$, which shows that $f|_{\{2027,2028,\ldots\}}$ is a non-decreasing function. For any $m \geq 2027$, note that $f(m)$ is a positive integer. Observe that for any $m \geq 2027$, if

$$f(m), f(m+1), f(m+2), \ldots, f(m+f(m))$$

are all equal, then $f(m + f(m) + 1)$ is greater than or equal to $f(m) + 1$. This shows that

$$f(m) \geq 1, \quad f(m) < f(m + f(m) + 1)$$

hold for any $m \geq 2027$. This shows that f takes arbitrarily large values.

Claim — Let $r \geq 2027$ be an integer such that

$$f(r) > \max\{f(1), f(2), \ldots, f(r-1)\}.$$

Then f takes equal values at the integers

$$r, r+1, r+2, \ldots, r+f(r)$$

and $f(r + f(r) + 1) = f(r) + 1$.

Proof of the Claim. The maximum number of times that a number appears in the list $f(1), f(2), \ldots, f(r-1)$ is equal to $f(r)$. Using the inequality $f(r) > \max\{f(1), f(2), \ldots, f(r-1)\}$, we see that any such number is less than $f(r)$. Therefore, the numbers $f(r), f(r+1), \ldots, f(r+f(r))$ are equal. It follows that $f(r + f(r) + 1) = f(r) + 1$. □

Since f takes arbitrarily large values, there exists an integer $r \geq 2027$ such that $f(r) > \max\{f(1), f(2), \ldots, f(r-1)\}$. Using the Claim and applying induction, it follows that the images of $r, r+1, r+2, \ldots$ under f are equal to

$$\underbrace{t, t, \ldots, t}_{t+1 \text{ times}}, \underbrace{t+1, t+1, \ldots, t+1}_{t+2 \text{ times}}, \underbrace{t+2, t+2, \ldots, t+2}_{t+3 \text{ times}}, \ldots$$

respectively, where $t = f(r)$. This implies that $f(n) = f(n + f(n))$ for infinitely many $n \in \mathbb{N}$, as desired. ■

Exercise 3 (INMO 2026 P1,). Let $x_1, x_2, x_3, \ldots$ be a sequence of positive integers defined as follows: $x_1 = 1$ and for each $n \geq 1$, we have

$$x_{n+1} = x_n + \lfloor \sqrt{x_n} \rfloor.$$

Determine all positive integers m for which $x_n = m^2$ for some $n \geq 1$. (Here $\lfloor x \rfloor$ denotes the greatest integer less or equal to x for every real number x.)

Walkthrough —
(a)

Solution 3. Let us establish the following claim.

Claim — For any integer $m \geq 2$, the inequalities

$$(m+r)^2 < m^2 + m + 2m + 2(m+1) + \cdots + 2(m+r-1),$$
$$m^2 + m + 2m + 2(m+1) + \cdots + 2(m+r-1) + m + r < (m+r+1)^2$$

hold for any integer r with $1 \leq r \leq m-1$.

Proof of the Claim. Note that

$$\begin{aligned} 2^2 + 2 + 2 \cdot 2 &= 10 \\ &> (2+1)^2, \\ 2^2 + 2 + 2 \cdot 2 + 2 + 1 &= 13 \\ &< (2+1+1)^2 \end{aligned}$$

holds, which shows that the claim is true for $m = 2$. Assume that the claim is true for some integer $m \geq 2$. Then, for any integer r with $1 \leq r \leq m-1$, we have

$$\begin{aligned} &(m+1)^2 + (m+1) + 2(m+1) + 2(m+1+1) + \cdots + 2(m+1+r-1) \\ &= m^2 + m + 2m + 2(m+1) + \cdots + 2(m+r-1) + 2m + 1 + 1 + 2r \\ &> (m+r)^2 + 2m + 2r + 2 \quad \text{(by the induction hypothesis)} \\ &> (m+r+1)^2, \end{aligned}$$

and

$$\begin{aligned} &(m+1)^2 + (m+1) + 2(m+1) + 2(m+1+1) \\ &\quad + \cdots + 2(m+1+r-1) + (m+1+r) \\ &= m^2 + m + 2m + 2(m+1) + \cdots + 2(m+r-1) + m + r \\ &\quad + 2m + 1 + 1 + 2r + 1 \\ &< (m+r+1)^2 + 2m + 2r + 3 \quad \text{(by the induction hypothesis)} \\ &< (m+r+2)^2. \end{aligned}$$

Moreover, the case $r = m$ can be verified directly as follows:

$$\begin{aligned} &(m+1)^2 + (m+1) + 2(m+1) + 2(m+1+1) \\ &\quad + \cdots + 2(m+1+m-1) \end{aligned}$$

$$
\begin{aligned}
&= (m+1)^2 + (m+1) + (3m+1)m \\
&= m^2 + 2m + 1 + m + 1 + 3m^2 + m \\
&= 4m^2 + 4m + 2 \\
&> (m+1+m)^2,
\end{aligned}
$$

and

$$
\begin{aligned}
&(m+1)^2 + (m+1) + 2(m+1) + 2(m+1+1) \\
&\quad + \cdots + 2(m+1+m-1) + (m+1+m) \\
&= 4m^2 + 4m + 2 + (2m+1) \\
&= 4m^2 + 6m + 3 \\
&< (m+1+m+1)^2.
\end{aligned}
$$

This completes the induction step, and hence the proof of the claim. □

Consider the map $f\colon \mathbb{Z}_{\geq 1} \to \mathbb{Z}_{\geq 1}$ defined by

$$f(k) = k + \lfloor \sqrt{k} \rfloor.$$

Let $m \geq 2$ be a positive integer. Consider the sequence defined by $y_1 = m^2$ and $y_{n+1} = f(y_n)$ for each $n \geq 1$. Note that

$$
\begin{aligned}
y_2 &= m^2 + m, \\
y_3 &= m^2 + m + m
\end{aligned}
$$

holds. Using $m^2 < y_3 < (m+1)^2$, and applying the Claim, it follows for any integer r with $1 \leq r \leq m-1$ that

$$y_{3+2r-1} = m^2 + m + 2m + 2(m+1) + \cdots + 2(m+r-1),$$

and

$$y_{3+2r} = m^2 + m + 2m + 2(m+1) + \cdots + 2(m+r-1) + m + r.$$

Since $m \geq 2$, applying the Claim, we see that

$$y_4, y_5, \ldots, y_{2m+1}$$

are not perfect squares. Note that y_2, y_3 are not perfect squares. This shows that none of

$$y_2, y_3, \ldots, y_{2m+1}$$

is a perfect square. Moreover, we have

$$
\begin{aligned}
y_{2m+2} &= y_{2m+1} + (m + m - 1) \\
&= m^2 + m + 2m + 2(m+1) + \cdots + 2(m+m-2) + m + m - 1
\end{aligned}
$$

The content posted here and at this blog by Evan Chen are quite useful.

$$
\begin{aligned}
&+ (m + m - 1) \\
&= m^2 + m + (3m - 2)(m - 1) + 4m - 2 \\
&= (2m)^2,
\end{aligned}
$$

which is a perfect square. Therefore, starting from any perfect square m^2 with $m \geq 2$, we reach the next perfect square $(2m)^2$ through repeated application of the map f, without encountering any other perfect square in between.

Note that

$$x_1 = 1^2, x_2 = 2, x_3 = 2 + 1 = 3, x_4 = 3 + 1 = 4 = 2^2.$$

Thus, the terms of the sequence (x_n) that are perfect squares are precisely the squares of the non-negative powers of 2. In other words, the positive integers m for which $x_n = m^2$ for some $n \geq 1$ are precisely the non-negative powers of 2. ■

References

[**Che25**] Evan Chen. *The OTIS Excerpts*. Available at https://web.evanchen.cc/excerpts.html. 2025, pp. vi+289 (cited p. 1)